31/03/2026

Tut - 10 (Boosting):

Sample	feature 1 (Age)	feature 2 (Income)	feature 3 (Credit Score)	Target (Loan Approved)
S1	25	40	650	Yes (+1)
S2	30	60	700	Yes (+1)
S3	45	35	600	No (-1)
S4	50	30	580	No (-1)
S5	28	55	720	Yes (+1)

Solution:-

Decision Stump

↓ Many

AdaBoost

1) Initial weights

$$w_i = \frac{1}{N} = \frac{1}{5} = 0.2$$

Predict $(h_1(x_i))$	w_i
+1	0.2
+1	0.2
-1	0.2
-1	0.2
+1	0.2

2) Decision Stump

Age < 40 → Yes (+1)

≥ 40 → No (-1)

3) Weighted Error

$$\boxed{\varepsilon_1 = \sum w_i \cdot \mathbb{I}(y_i \neq h_1(x_i))}$$

$$= [0.2\times0 + 0.2\times0 + 0.2\times0 + 0.2\times0 + 0.2\times0]$$

$$= 0.$$

4) Classifiers / Model Error weight:

$$\alpha_1 = \frac{1}{2}\log\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$$

$$= \frac{1}{2}\log\left(\frac{1-0}{0}\right) \rightarrow \infty$$

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5) Update weights :

$$w_i^{New} \rightarrow w_i e^{-y_i \alpha_1 h_1(x_i)}$$

$$\downarrow$$

$$e^{-\infty} = 0.$$

6) Prediction

Model will stop training

$$H(x) = h_1(x_i)$$

If S5 - No. - +1.

3. Weighted error.

$$\varepsilon_1 = \sum w_1 \cdot I(y_i \neq h_1(x_1))$$

$$= 0.2 \times 1 = 0.2.$$

4) Classifiers / Model Error :

$$\alpha_1 = \frac{1}{2} \log\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$$

$$= \frac{1}{2} \log\left(\frac{1-0.2}{0.2}\right) = 0.693$$

5) Update weights :

$$w_i^{New} \rightarrow w_i e^{-y_i \alpha_1 h_1(x_i)}$$

$$= w_1 0.2 \times e^{-1 \times 0.693 \times 1}$$

$$= 0.1.$$

* Normalized w.

$$= \frac{w_i}{\sum w_i}$$

w^N
0.125
0.125
0.125
0.125
0.5

6) Prediction

$$H(x) = \text{sign}\left[\sum_{t=1}^{T} \alpha_t h_t(x_i)\right]$$

$$= \text{sign}\left[\alpha_1 h_1(x_i) + \alpha_2 h_2(x_2) + \cdots\right]$$